

stores energy-efficient, and provide means for sustainable end-of-life product disposal.

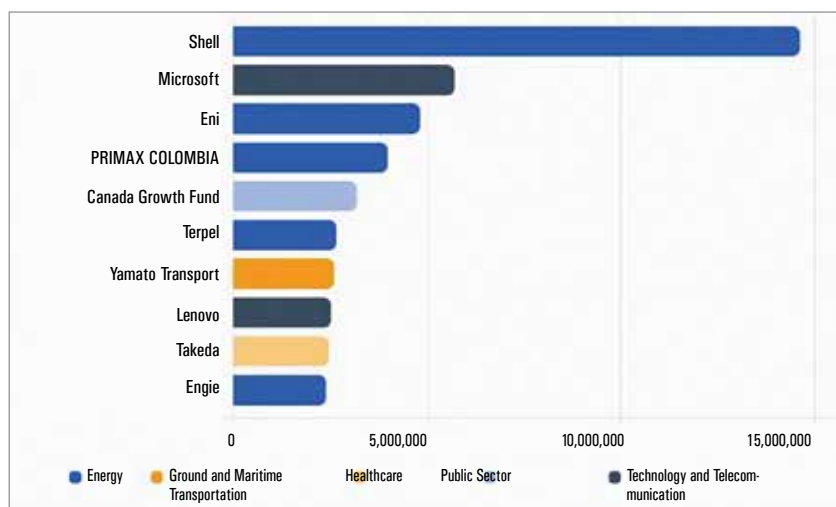
- **Invest in carbon credits and offsets:** Purchase carbon credits and invest in engineering-based and nature-based carbon removal technologies to offset your emissions.

## Understanding carbon credits and offsets

Carbon credits are created or generated by projects that reduce or remove greenhouse gases from the environment. There are engineering-based projects such as direct air capture, methane capture, and renewable energy generation, as well as nature-based projects such as reforestation and soil sequestration. One carbon credit is generated when one metric tonne of GHGs is removed or reduced from the environment. These projects and their results are verified and quantified (into carbon credits) by government-accredited agencies. And so, one carbon credit eventually becomes a tradeable permit to emit one metric tonne of carbon dioxide or other emissions. When a company uses a carbon credit to offset its emissions, the credit is said to be retired and cannot be traded in the market again.

There are two ways in which the carbon market may be regulated: the cap-and-trade method or the baseline-and-credit approach.

- **Cap-and-trade:** In the cap-and-trade approach, the government sets an overall cap, or upper limit,



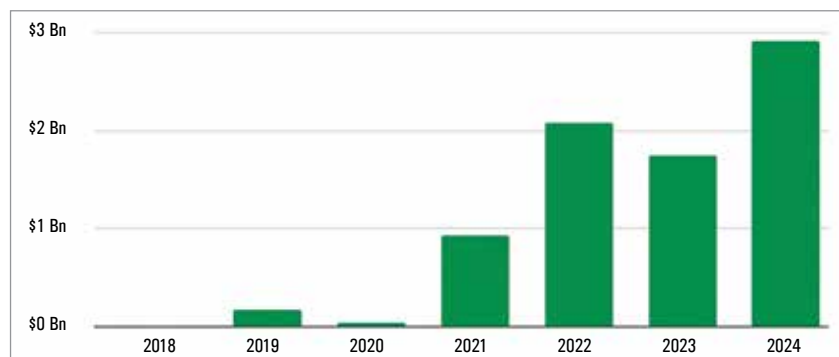
Top buyers of carbon credits in 2024 (Source: Allied Offsets)

on permitted emissions for each regulated sector/industry. This is divided into smaller allowances or credits and distributed to the different entities in that sector, either through direct allocation or an auction. If a company emits less than its allowance, it can sell the surplus credits to others who have overshot their limits. The cap must be lowered periodically to ensure that companies work towards reducing their emissions. The cap-and-trade system incentivises emissions reductions, as companies can sell the saved carbon credits for profit. Tesla, for example, generated a \$2.76 billion revenue in 2024 from the sale of carbon credits—a 54% year-on-year

increase from 2023. The cap-and-trade approach has been adopted by many countries, including China, New Zealand, parts of the United States, the United Kingdom, and the European Union.

- **Baseline-and-credit:** The other approach, baseline-and-credit, sets a predefined baseline for each regulated activity or industry, based on a study of average emissions using standard technology or practices. Entities that emit less than the baseline earn carbon credits, which they can sell to those that overshoot the baseline. Unlike the cap-and-trade system, there is no cap on total emissions for the industry or sector here. Some experts feel this may be ineffective in achieving the overall net-zero goals of a nation, because the overall emissions by the sector are unpredictable and often uncontrolled. The baseline-and-credit approach is adopted by some countries, like India. Some, like Australia, follow a hybrid approach, a baseline-and-credit system with an overarching cap on the total emissions.

- **The ugly side of carbon credits:** It has been noted that companies that really need to make substantial



Private investment in carbon dioxide removal technologies continues to be on the rise, clocking nearly \$3 billion in 2024 (Source: AlliedOffsets)